BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

Name:

ID:

> **Due Date**
>
> Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\holly\4750 HW\hw3-ellasandwich`
```

```julia
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day-1, and the decay rates of CBOD and NBOD are are 0.35 and 0.25 day-1, respectively. The river's saturated dissolved oxygen concentration is 10m g/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance `d` km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m3/d	10,000 m3/d	15,000 m3/d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10

C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
# first 15m, before 2nd discharge
```

```julia
for i in 0:15
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end

# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)
Cx = C_0
# remaining 35 m (m16-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
    Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
    push!(C_x, Cx)
end
min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")

### Discretized model
C_x_disc = []
dx = 0.5
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
Cx = C0
for i in 0:dx:15-dx
    push!(C_x_disc, Cx)
    C_x1 = Cx + (dx/U) * (ka*(Cs-Cx) - kc*B0*exp(-kc*i/U)-kn*N0*exp(-kn*i/U))
    Cx = C_x1
end
C_0 = (110000*C_x_disc[end]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)
Cx = C_0
for i in 15:dx:50
```

```
    x = i-15
    push!(C_x_disc, Cx)
    C_x1 = Cx + (dx/U) * (ka*(Cs-Cx) - kc*B_0*exp(-kc*x/U)-kn*N_0*exp(-kn*x/U))
    Cx = C_x1
end
min_Disc = minimum(C_x_disc)
println("Minimum DO, Numerical (mg/L): $min_Disc")
# graphing
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", yla
x2 = [0:dx:50]
plot!(x2, C_x_disc, color = "orange", label="Numerical Solution")
vline!([0,15], linestyle=:dot, label="Discharges", color = "grey")
display(p)
```

```
Minimum DO, Streeter-Phelps (mg/L): 3.757481773041718
Minimum DO, Numerical (mg/L): 3.6412522420697764
```

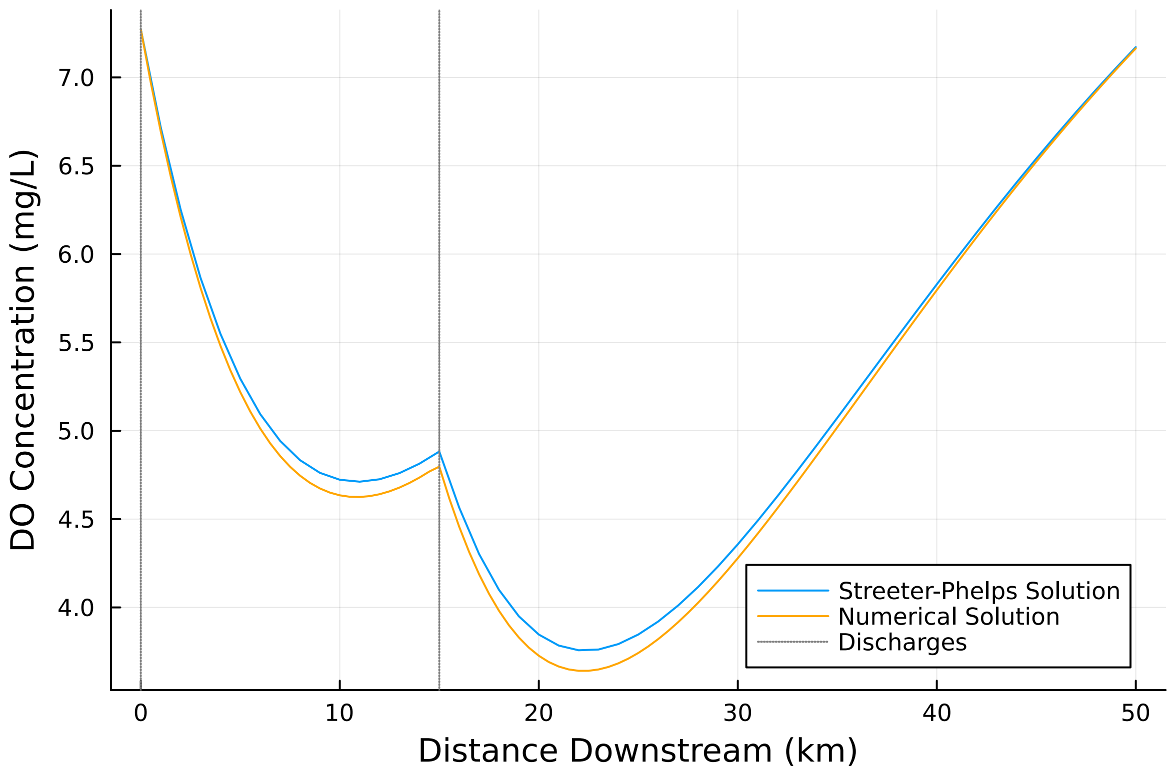


Figure 1: Analytical vs. Numerical solution, with the numerical being an underapproximation of the actual behavior

Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

The numerical solution starts the same and pretty much ends the same as the streeter-phelps, but in between, the values are slightly below that of the streeter-phelps. This being said, the minimum value for the discretized solution is slightly lower than the analytical solution (3.64 vs. 3.76). this is due in part to our relatively large timestep. When we lessen the timestep to 0.1, the solution is almost identical, so some of this error can be attributed to an inadequatey sized timestep. We believe it is lower because the slope of the tangent line at the beginning is negative, and remains fairly constant at the beginning, allowing our estimates to follow the curve pretty closely using the slope of the tangent line. As a point of inflection is approached, the slope of the tangent line drastically changes and it is hard for the model to keep up and adjust accordingly, especially with such a large timestep. This explains why the estimate is below the analytical solution, and in a similar manner, we expect that if the graph were to be concave, the numerical solution would be an over-approximation because the slope of the tangent line would start as steeply positive.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

```
# Replication for solving problem 1.3, only stream 1
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10
# percentage removal, through guess-and-check
p = 0.277
```

```
# Target DO: 4.2 mg/L

C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50*(1-p))/(110000)
N0 = (100000*5+10000*35*(1-p))/(110000)
# first 15m, before 2nd discharge
for i in 0:15
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end
# getting value at x=15 before discharge
a1 = exp(-ka*15/U)
a2 = (kc/(ka-kc))*(exp(-kc*15/U)-exp(-ka*15/U))
a3 = (kn/(ka-kn))*(exp(-kn*15/U)-exp(-ka*15/U))
Cx_15 = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)

# remaining 35 m (m16-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
    Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
    push!(C_x, Cx)
end
min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", yla
display(p)
```

```
Minimum DO, Streeter-Phelps (mg/L): 4.202505257381106
```

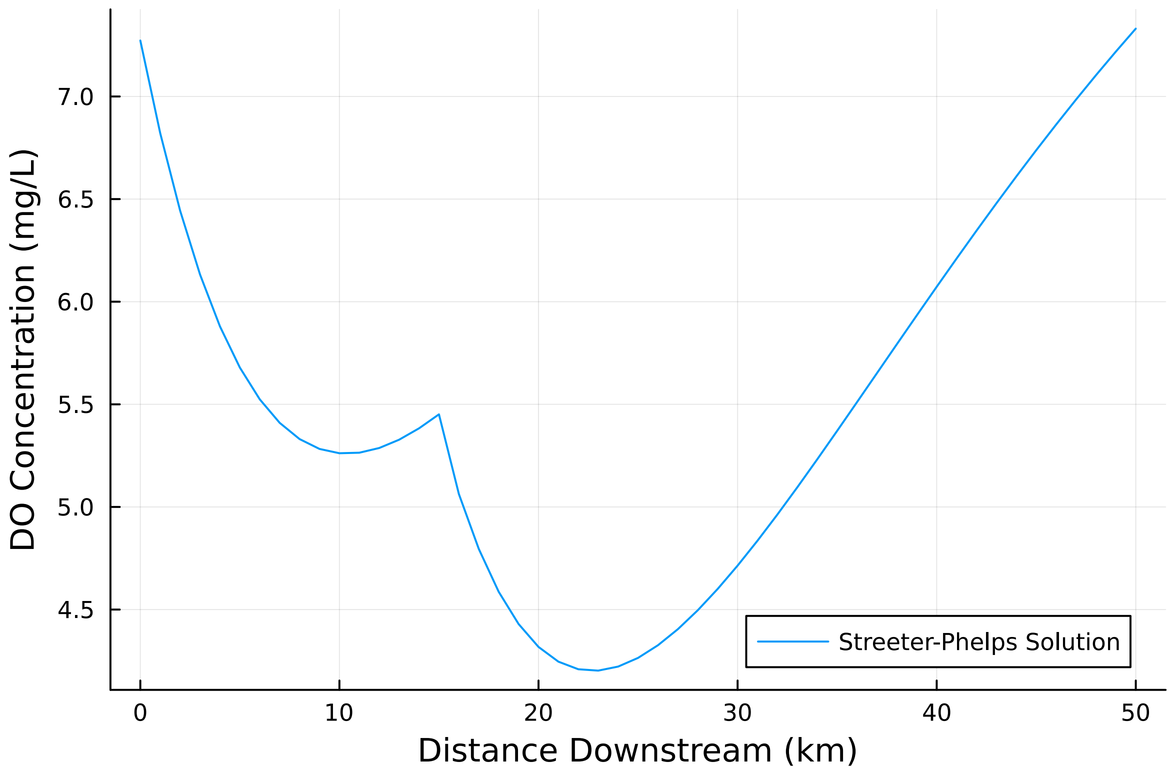


Figure 2: DO behavior only treating stream 1. Stays above 4.2mg/L at 27.7% removal.

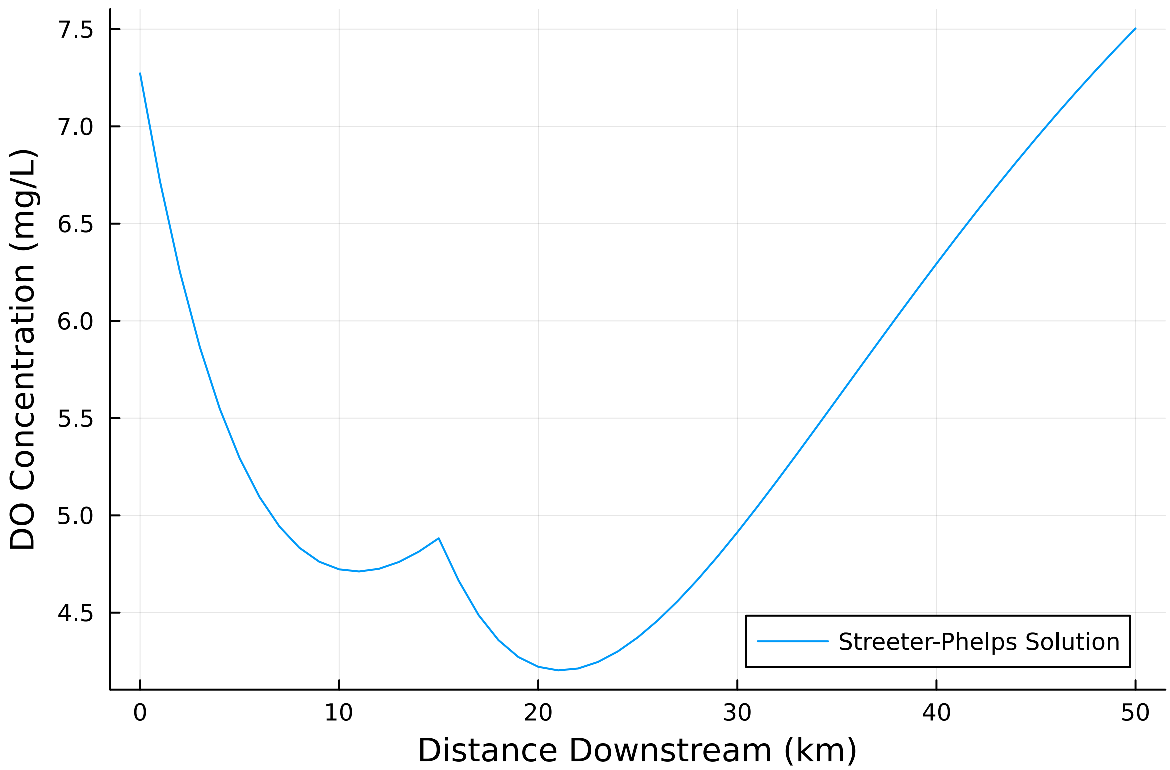
```
# Replication for solving problem 1.3, only stream 2
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10
# percentage removal
p = 0.219
# Target DO: 4.2 mg/L

C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
# first 15m, before 2nd discharge
for i in 0:15
```

```
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end

# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45*(1-p))/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35*(1-p))/(125000)

# remaining 35 m (m16-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
    Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
    push!(C_x, Cx)
end
min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", yla
display(p)
```




```
Minimum DO, Streeter-Phelps (mg/L): 4.203266648522453
```

Figure 3: DO behavior only treating stream 2. Stays above 4.2mg/L at 21.9% removal.

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

You would have to remove less waste of stream 2 to meet the criteria, but only by a small amount (~22% vs ~28%). Therefore, more information is needed to determine which plan to choose. Cost of treatment for each would be beneficial, so we could determine which is cheaper and therefore more attractive a choice. If treating stream 1 is cheaper, you would most likely want to either treat only stream 1 or both streams equally. If both are equivalent in cost, then just treating stream 2 would be better due to less removal and therefore less resources allocated.

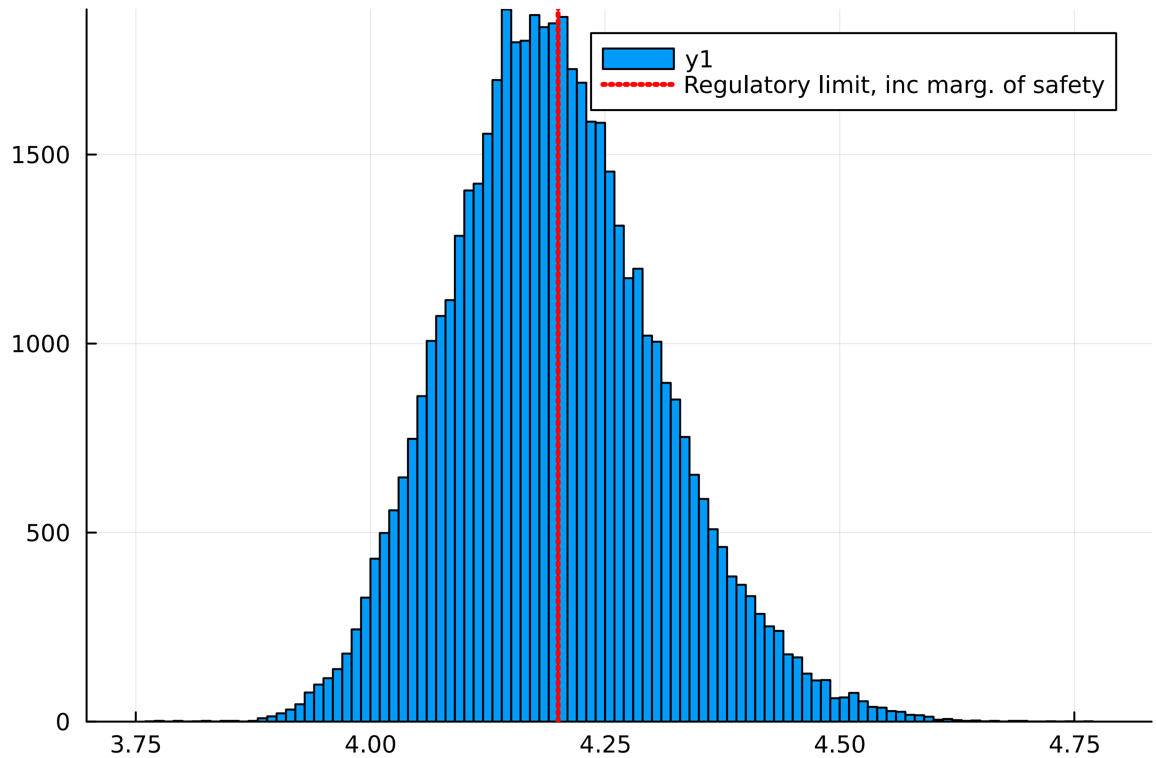
Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

```
# Replication for solving problem 1.5, Monte Carlo
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10
# percentage removal
p = 0.277
# Target DO: 4.2 mg/L
min_array = []
inflow = LogNormal(2, 0.15)
len = 50000
in = rand(inflow, len)

# Running Monte Carlo
for i in 1:length(in)
    C_x = []
    ### Analytic Model, initial C0,B0,and N0
    C0 = (100000*in[i]+10000*5)/(110000)
    B0 = (100000*5+10000*50*(1-p))/(110000)
    N0 = (100000*5+10000*35*(1-p))/(110000)
    # first 15m, before 2nd discharge
    for i in 0:15
        a1 = exp(-ka*i/U)
        a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
        a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
        Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
        push!(C_x, Cx)
    end
    # now, initial vars considering first discharge for 2nd
    C_0 = (110000*C_x[16]+15000*5)/(125000)
    B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
    N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)
```

```
    # remaining 35 m (m16-m50) after 2nd discharge
    for i in 16:50
        x = i-15
        a1 = exp(-ka*x/U)
        a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
        a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
        Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
        push!(C_x, Cx)
    end
    min_SP = minimum(C_x)
    push!(min_array, min_SP)
end
# Plot histogram as sanity check
x = [0:len]
ave = mean(min_array)
pl = histogram(x, min_array, bins=100)
vline!([4.2], linestyle=:dot, linewidth = 2.5, label="Regulatory limit, inc marg. of safety"
display(pl)
println("expected Value: $ave")
# Calculate Probability
no = 0
yes = 0
for i in 1:length(min_array)
    if min_array[i]<4.2
        no = no + 1
    else
        yes = yes + 1
    end
end
# probability
prob = no / (yes+no)
println("Probability of not meeting standard: $prob")
a = 0.05
ci = (quantile(min_array, a/2), quantile(min_array, 1 - a/2))
println("95% CI for random variable = $ci")
```




```
expected Value: 4.197443057976062
Probability of not meeting standard: 0.53314
95% CI for random variable = (3.998240738441207, 4.435831790030583)
```

Figure 4: Histogram showing the spread of Monte Carlo values of minimum DO after removing waste from stream 1 only.

I determined my size for the Monte Carlo model by starting low and testing it, looking at the resulting histogram and the expected value. Starting at 1000, the expected value was fluctuating between values and the histogram was quite different each time. Continuing to test and raise this value, I reached a size of 50,000, which takes around 6 seconds to run, so it is not too large. The histogram remains pretty consistent, and the expected value only varies slightly in between runs. Since this sample size does not include any costs on the model, such as run-time, it is sufficiently large and not too large.

References

List any external references consulted, including classmates.